

# LYRA-CPP — THETIS DIRECT-PORT PLAN (TX)

**Authoritative doc for the current TX work direction.**

*[!] The older EXECUTION\_PLAN.md (Step 14 wire-layer rebuild, Phase 3 etc.) describes a DIFFERENT approach that is now superseded for the TX work. Do NOT mix the two plans — that doc tracks an earlier wire-rebuild pass which was partial and Lyra-native-leaning. THIS doc is the active plan going forward. Old doc remains in tree for historical reference only (some shipped pieces from it are still in production — noted per-component in the registry below).*

**Branch:** tx-rebuild (the direct-port work lives here; origin/main carries the earlier Step 14 / TX-1 line and is deliberately untouched by the direct-port arc).

**Reference tree (the ONLY trusted TX reference):**

- D:\sdrprojects\OpenHPSDR-Thetis-2.10.3.13\Project Files\Source\
  - wdsp\\*.c — WDSP DSP engine (GPL v3+, port-with-attribution OK)
  - ChannelMaster\\*.c — wire layer (GPL v3+, port-with-attribution OK)
  - Console\\*.cs — C# UI/control glue (study ONLY; write Lyra-native equivalents)

- Y:\Claude local\\_hl2src\ — HL2+ ak4951v4 gateway RTL

(control.v / radio.v / usopenhpsdr1.v etc. — the ground truth when host-side reference and gateway disagree)

**Provenance & attribution rule (operator-locked 2026-06-09, supersedes the 2026-05-31 "no-attribution-in-commits" tightening):**

Lyra-cpp is a derivative work of openHPSDR Thetis (ChannelMaster + WDSP modules), GPL v3+. The GitHub project description openly states this: **"TX baseline ported from openHPSDR Thetis (ChannelMaster) with Lyra-native DSP additions."** Open attribution is both legally required (GPL) and operator-intended (public project posture).

1. **License attribution at file head — REQUIRED.** Any file ported from WDSP or ChannelMaster carries a header block crediting upstream (NR0V for WDSP modules; openHPSDR Thetis for ChannelMaster modules) + the GPL v3+ notice. Existing AAMix.{h, cpp}, CMaster.cpp, RadioNet.cpp etc. already follow this; keep it.
2. **Code comments citing source — ENCOURAGED.** Citations may name the reference openly (Reference: Thetis ChannelMaster cmaster.c:297-313, // Per openHPSDR networkproto1.c:1078, // HL2+ ak4951v4 gateway control.v:209-220). The file:line is the primary technical content; the reference-app name is context that helps future readers find the source tree.
3. **Commit messages — REFERENCE-APP NAMES PERMITTED.** A commit like **"port aamix.c from Thetis ChannelMaster with Lyra-native idiom translations"** is fine — it matches the public GitHub posture and makes the history grep-able. The earlier 4cef88d tightening (2026-05-31) that put commit messages on the no-attribution list is **rescinded**.
4. **\*\*Public docs (README, docs/\*.md design docs, memory files, this plan) — OPEN ATTRIBUTION.\*\*** Same posture as the GitHub description.
5. **User-facing UI strings — KEEP OPERATOR-FOCUSED.** UI labels operators see in normal operation (panel titles, button text, error toasts) describe Lyra behavior, not provenance. **"AAMix output (HL2 jack)"** not **"Thetis-porting AAMix output"**. About dialog / credits screen / install docs MAY (and probably should) name openHPSDR Thetis + WDSP/NR0V openly — credit where due. This is the only category that stays "Lyra-native voice in operator-facing labels," because operators interact with Lyra, not its source provenance.
6. **The "Lyra-Native style governs surrounding architecture only" rule (Stage B-locked) STANDS unchanged.** Reference DSP API call patterns are byte-for-byte faithful; Qt/Vulkan/QML/process model is Lyra-native.
7. **Reference precedence STANDS unchanged.** Verify TX ONLY against Thetis 2.10.3.13 (D:\sdrprojects\OpenHPSDR-Thetis-2.10.3.13\...) + the HL2+ ak4951v4 gateway RTL (Y:\Claude local\\_hl2src\). **Old**

**Python Lyra is NOT a trusted TX reference** (its TX never worked).

**Consequence of the 2026-06-09 amendment:** the three historical commits (efdde02, 604d87b, ed9f0cd) that "predated the no-attribution rule" and contained "Thetis" in their messages are **no longer in violation** — they're now consistent with the rule. Task #73 ("Cleanup pre-existing 'Thetis' mentions in shipped code") is **OBSOLETE**d by this amendment.

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## LOCKED METHODOLOGY (Stage B-validated, do NOT relax)

These rules were earned through the Stage B aamix.c port arc (2026-06-08) — see SESSION\_LOG.md for the full trail.

1. **\*\*Smallest revertable empirical step -> operator HL2 bench ->**

next step.\*\* Each commit is one focused change with a falsifiable bench gate. Multi-step "design then implement" is for stages that genuinely need it (operator approves); default is iterate-and-bench.

2. **"Do as Thetis does, Lyra-Native style"** (locked

2026-05-30 after the fexchange0 crash arc, re-validated tonight after the B.6.b shortcut attempt cost a revert cycle). Three sub-rules:

- Every WDSP API call matches the reference byte-for-byte

(parameter values, order, the SET of setters fired, the lifecycle stage at which they fire). If tempted to pick a value the reference doesn't use, STOP and find what the reference uses. "I have a reason to pick differently" means I don't. Reference has been right every time.

- "Lyra-Native style" governs surrounding architecture ONLY

(Qt + Vulkan + QML + process/thread model + facade APIs + IPC + UI shell). The WDSP DSP-engine call surface is reference-bound.

- Provenance only in code COMMENTS + design docs + memory.

Never commit messages, never UI strings.

3. **Counters first when the symptom is "nothing happens".**

The Stage B 2-stage instrumentation (chain counters in xMixAudio/mix\_main + dispatch counters in dispatchAudioFrame) localized the silent-audio defect to a single line of code on the 2nd bench. Pure speculation would have cost a multi-round revert/speculate cycle. Atomic counter + rate-limited qInfo = trivial overhead, very high diagnostic ROI when "I press play, hear nothing, log looks normal."

4. **Operator-empirical rule overrides agent inference.**

When operator hardware data contradicts what an audit/red-team/Plan-agent concluded, the data wins. Do not defend the audit — re-open, bisect, fix. This rule has fired repeatedly through the broader project arc; the Stage B B.6.b-fix1 fix was an instance (chain counters healthy suggested AAMix was correct; the temptation was to revert the whole port; the empirical pattern instead pointed at the wire-up surface where the bug actually was).

5. **\*\*Read the reference FIRST, in shipped-code form, before**

designing.\*\* Every time we designed first and read second, we shipped a regression. Every time we read first, we got the right answer in one pass. The no-attribution rule applies to shipped code; it does NOT mean "don't read the reference." Read it, dossier the file:line in the plan doc, then write Lyra-cpp.

6. **Session-start discipline:** git fetch origin is the

FIRST action every session, before reading any local state. If divergence exists, surface it to the operator BEFORE writing code. No exceptions.

7. **Push after every component ships, not at EOD.** Each

clean operator HL2 bench -> backup bundle + push immediately. Smaller in-flight delta = smaller surprise blast radius.

8. **Backup bundle naming:** \_backups/lyra-cpp-YYYY-MM-DD-.bundle.

git bundle create ... --all (covers all refs).

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## STATUS REGISTRY — what's done, in-progress, pending

Format: per-file or per-component row. Status tags (ASCII so they render identically in the PDF/DOCX siblings):

- [DONE] DONE + operator-bench-validated
- [WIP] IN-PROGRESS (committed but not yet bench-validated, or partially shipped)
- [PEND] PENDING (planned, not started)
- [N/A] N/A (Lyra-native equivalent already shipped; no direct port planned — see notes column)

### Wire layer (ChannelMaster — port directly w/ attribution)

| Source                                                            | Target (lyra-cpp)                      | Status | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------|----------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WDSP exports linkage (OpenChannel/fexchange0/analyzers/PS family) | src/wire/wdspcalls.{h,cpp} (P0.a)      | [DONE] | <b>P0.a</b> e0c0f4a <b>2026-06-09</b> . The single operator-approved linkage seam: fn-ptr table carrying the WDSP exports' exact names, resolved once from the loaded wdsp.dll. PureSignal entry family complete (verified vs dumpbin). RESAMPLE-typed + flush_resample added at P0.c.                                                                                                                                                                         |
| ChannelMaster\cmcomm.h (family common header)                     | src/wire/cmcomm.{h,cpp} (P0.b -> P0.d) | [DONE] | complex typedef + PORT + verbatim malloc0 (wdsp/utilities.c:37-43) + PI + the reference include surface. P0.d added the opaque verbatim twin typedefs for deferred-subsystem types (ANB/NOB/VOX/TXGAIN/ANALYZERS + DELAY at P2.a); <b>EER completed at P2.a 2026-06-12</b> — full verbatim struct (wdsp/eer.h:30-49) + the umbrella-mapping note (.cpp files include explicit family headers — an include-list umbrella would be circular under #pragma once). |
| ChannelMaster\ilv.c (TX I/Q interleaver)                          | src/wire/ILV.{h,cpp} (P0.b)            | [DONE] | <b>P0.b</b> 1f49d98 <b>2026-06-11</b> . VERBATIM direct port (ilv, *ILV twin typedef, raw Outbound fn ptr, pcm->xmtr[].pilv setters, malloc0/_aligned_free). Mechanical diff vs ilv.{h,c} = whitespace-only. The earlier src/wdsp/ILV retrofit + its pilv[] bank are retired. scratch/test_ilv ALL PASS.                                                                                                                                                       |
| ChannelMaster\cmbuffs.{h,c} (CMB ring + cm_main pump)             | src/wire/CmBuffs.{h,cpp} (P0.d)        | [DONE] | <b>P0.d</b> afc7950 <b>2026-06-12</b> . VERBATIM rewrite (cmb, *CMB, #define CMB_MULT (3), malloc0/_aligned_free; the Phase-C retrofit's calloc/intptr_t/guard deviations removed; pcm->in[] alloc moved back to create_cmastar per reference). Mechanical diff = whitespace-only. [DONE] <b>P0.d operator HL2 RX-regression bench PASSED 2026-06-12</b> .                                                                                                     |
| ChannelMaster\cmsetup.{h,c} (radio structure + id helpers)        | src/wire/cmsetup.{h,cpp} (P0.d, NEW)   | [DONE] | <b>P0.d</b> afc7950. VERBATIM: cmMAX* sizing macros (16/4/4/2/32), rxid/txid/sp0id/stype/chid/inid/mixinid/getbuffsize, SetRadioStructure + set_cmdefault_rates. main.cpp config: SetRadioStructure(2,1,1,1,0,...) -> chid(0,0)=0 RX1 / chid(1,0)=1 TX matches the live channel layout. [DONE] Bench PASSED 2026-06-12.                                                                                                                                        |

| Source                                                                              | Target (lyra-cpp)                                               | Status | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ChannelMaster\cmaster.{h,c}<br>(FULL: struct + create/destroy + xcmaster + setters) | src/wire/CMaster.{h,cpp}<br>(P0.d)                              | [DONE] | <b>P0.d</b> afc7950. VERBATIM rewrite: full _cmaster struct (cmaster.h:39-99 incl. the PS surface — out[3]/panalalloc/pgain/peer), cmaster, *CMaster, raw TCI callback fn ptrs, enum AudioCODEC, cm = {0}. Verbatim no-arg create_xmtr/destroy_xmtr (out[0..2] malloc0 + OpenChannel(ch 1) + XCreateAnalyzer(TX disp 1) + create_ilv through the wdspcalls seam — <b>the TxChannel RAIL carve-out is DELETED</b> , src/wdsp/TxChannel.{h,cpp} retired). create_cmaster/destroy_cmaster verbatim per-stream loops (update[] CS + cmbuffs + in[] for all cmSTREAM streams). xcmaster verbatim (update[] CS restored, real stype/txid/chid, TCI memset restored; fexchange0 + monitor xMixAudio + xilv live). All Sendp*/Set* setters ported. Deferred subsystem lines (dexp/anti-vox/txgain/sidetone/pipe/cmasio/analyzers.c/create_rcvr-body) carried IN PLACE as reference text with DEFERRED tags; <b>eer RESTORED VERBATIM at P2.a 2026-06-12</b> (create_eer run=0 / destroy_eer / xeer / pSetEER* via the wdspcalls seam — the object exists so the sendOutbound/sendProtocol1Samples peer->run derefs are valid, per cmaster.c:212-224). Sole code accommodation: (char*)" at XCreateAnalyzer. <b>[DONE] Operator HL2 RX-regression bench PASSED 2026-06-12</b> (RX working on the new per-stream pump/ring layout). |
| ChannelMaster\obbuffs.c (TX-out ring/seam)                                          | src/wire/ObBuffs.{h,cpp}<br>(P1, NEW)                           | [WIP]  | <b>P1 SHIPPED 2026-06-12</b> — obbuffs.{h,c} verbatim (obb/*OBB twin typedef, file-scope obp four-alias bank, the reference's own calloc/free 2014-era idioms kept). Mechanical diff: .cpp IDENTICAL; .h sole delta = the include-guard define replaced by #pragma once (documented packaging). Sole deferred line: sendOutbound(id, a->out) in ob_main (network.c:1237 = P2 scope), carried as reference text. TU is DORMANT — no Lyra caller of create_obbuffs until P3/P4; wire bytes unchanged.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| ChannelMaster\networkproto1.c (HL2 wire writer)                                     | src/hl2_stream.cpp (LIVE) + src/wire/* (Step 14 pieces)         | [WIP]  | This is the file the Step 14 plan was attacking. Currently a MIX of Step 14 Lyra-native rewrites + the original hl2_stream.cpp body. P2 = sendOutbound/sendProtocol1Samples fidelity audit of the dormant wire layer; P4 = Wire-LIVE switchover (one commit, full HL2 bench gate).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| ChannelMaster\network.c (UDP + keepalive)                                           | src/wire/Ep2SendThread.{h,cpp} + src/wire/Ep6RecvThread.{h,cpp} | [WIP]  | Step 14 Lyra-native rewrites shipped (\$5/\$6); could be re-reported direct from reference, OR left as-is if reference-parity audit passes. Operator call (folds into P2/P4).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Source                                        | Target (lyra-cpp)                         | Status | Notes                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------|-------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ChannelMaster\netInterface.c:<br>:create_rnet | src/wire/RadioNet.cpp::create_rnet (LIVE) | [DONE] | Direct port shipped earlier. The Stage-A SendpOutboundRx no-op stub at line 272 was DELETED 2026-06-08 (B.6.b-fix1, commit 8b8e0da) because it clobbered the AAMix wire-up; future codec-id switching must NOT re-introduce a default registration here. P3 = the outbound registrations (SendpOutboundTx(OutBound) etc.) — MUST register AFTER create_xmtr (the verbatim setters have NO null guards, reference ordering). |

### WDSP RX chain (port directly w/ attribution)

| Source                       | Target (lyra-cpp)             | Status | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------|-------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ChannelMaster\aamix.{c,h}    | src/wire/AAMix.{h,cpp} (P0.c) | [DONE] | <b>P0.c</b> e0f2584 <b>2026-06-11 — VERBATIM re-port</b><br>(supersedes the 2026-06-08 Stage-B C++23-idiom translation, which the operator rejected as a deviation class).<br>aamix, *AAMIX twin typedef + anonymous slew struct, raw Outbound fn ptr, paamix[] bank private to the .cpp, Win32 primitives direct. Mechanical diff vs aamix.{h,c} = whitespace-only.<br>src/wire/resample.h = the public RESAMPLE ABI verbatim. WdspEngine RX wire-up: capturing lambda -> static member fn + TU-scope context ptr (the reference free-fn+global shape). [DONE]<br><b>Operator HL2 RX bench PASSED 2026-06-11.</b> |
| wdsp\firmin.c (filter cores) | (consumed via WDSP cffi)      | [N/A]  | Used internally by WDSP RXA chain via cdef + dlsym. No source-level port needed (the DLL ships with Lyra-cpp via _native/ per the Python-project pattern; cdef + dlsym in wdsp_native.cpp). WISDOM cache fix shipped on Python side; lyra-cpp's RX chain inherits the same path.                                                                                                                                                                                                                                                                                                                                   |
| wdsp\wisdom.c                | (consumed via WDSP cffi)      | [N/A]  | Same as firmin.c.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### WDSP TX chain (per CLAUDE.md §4.1 port table)

| Source                            | Target (lyra-cpp)                              | Status | Notes                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------|------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| wdsp\TXA.c (channel scaffold)     | (consumed via wire/wdspcalls seam — P0.a/P0.d) | [DONE] | <b>TxChannel class DELETED at P0.d</b><br>afc7950 — the verbatim create_xmtr in wire/CMaster.cpp opens the WDSP TXA channel itself (OpenChannel/fexchange0/CloseChannel through the wdspcalls seam, cmaster.c:177-190 byte-exact), which removed the runtime-DLL RAIL justification the class existed for. wdsp_native.{h,cpp} still carries the SetTXA*/GetTXAMeter cdefs for the operator-control surface. |
| wdsp\bandpass.c (bp0/bp1)         | (consumed via WDSP cffi — SetTXABandpassFreqs) | [N/A]  | The §15.23 trap (SetTXABandpassRun overrides bp1 to a stale state) is documented in CLAUDE.md + project_lyra_cpp_tx.md. cffi-only access is the correct posture.                                                                                                                                                                                                                                             |
| wdsp\compress.c (TX compressor)   | src/wdsp/Compress.{h,cpp} OR cffi binding      | [PEND] | Operator choice: direct port (~150 LOC) for v0.2.1 EQ + dynamics ship, OR cffi-only binding if WDSP DLL exports are sufficient. Direct port recommended for parity with Lyra's existing UI control surface.                                                                                                                                                                                                  |
| wdsp\cfcomp.c (5-band CFC)        | src/wdsp/CFComp.{h,cpp} OR cffi binding        | [PEND] | Same operator-choice. ~600 LOC; one of the heavier ports. Replaced in Python Lyra plan by a Lyra-native Combinator per §15.19; cpp may take the same path (custom Combinator instead of WDSP CFC) — operator design call.                                                                                                                                                                                    |
| wdsp\osctrl.c (CESSB)             | cffi binding (SetTXAosctrlRun)                 | [PEND] | Likely cffi-only; ~200 LOC port not justified unless operator UI need surfaces.                                                                                                                                                                                                                                                                                                                              |
| wdsp\wcpagc.c mode 5 (leveler)    | cffi binding (SetTXALeveler*)                  | [WIP]  | cdef already present in wdsp_native.cpp. UI surface exists in TX-1 components on main. Confirm reference-parity at the call-site setter sequence.                                                                                                                                                                                                                                                            |
| wdsp\wcpagc.c mode 5 (ALC)        | cffi binding (SetTXAALCMaxGain)                | [WIP]  | cdef present; SetTXAALCThresh does NOT exist (§15.23-class trap — only MaxGain governs ALC ceiling).                                                                                                                                                                                                                                                                                                         |
| wdsp\eqp.c (parametric EQ)        | src/wdsp/EQ.{h,cpp} OR cffi binding            | [PEND] | Per CLAUDE.md §4.1 v0.2.1; ~300 LOC. Lyra-native parametric EQ likely (§15.19 lists 3-or-5-band parametric replacing the WDSP 10-band graphic).                                                                                                                                                                                                                                                              |
| wdsp\gen.c (gen0/gen1 generators) | cffi binding (SetTXAPreGen*/SetTXAPostGen*)    | [WIP]  | gen1 (postgen) used today for TUN tone. gen0 (pregen) cdefs deferred per §15.23 (bench-tooling nicety, not on the signal path).                                                                                                                                                                                                                                                                              |

### ***TX PureSignal (per CLAUDE.md §4.1 v0.3)***

| Source                 | Target (lyra-cpp)                      | Status | Notes                                                                                                  |
|------------------------|----------------------------------------|--------|--------------------------------------------------------------------------------------------------------|
| wdsp\iqc.{c,h}         | src/wdsp/Iqc.{h,cpp} (~315 LOC)        | [PEND] | v0.3. cffi math + Lyra-cpp wrapper for the 5-state PS lifecycle (RUN, BEGIN, SWAP, END, DONE).         |
| wdsp\calcc.{c,h}       | src/wdsp/Calcc.{h,cpp} (~1164 LOC)     | [PEND] | v0.3. cffi math + Lyra-cpp wrapper for the 8-state PS FSM + 3-state auto-attenuator FSM + PSDialog UI. |
| wdsp\lmath.c::xbuilder | src/wdsp/PsXBuilder.{h,cpp} (~200 LOC) | [PEND] | v0.3. Cubic-spline coefficient builder, used by Lyra-cpp-side calcc orchestration.                     |
| wdsp\delay.c           | src/wdsp/DelayLine.{h,cpp} (~80 LOC)   | [PEND] | v0.3. TX/feedback time-alignment.                                                                      |

### Console (C# — study only, Lyra-native equivalents)

| Source                                     | Target (lyra-cpp)                                   | Status | Notes                                                                                                                                                                                                                                                                                  |
|--------------------------------------------|-----------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Console\console.cs::chkMOX_CheckedChanged2 | src/ptt.{h,cpp} + src/wdsp_engine.cpp keydown/keyup | [WIP]  | Lyra-native FSM shipped on main. Verified against the reference's keydown/keyup ordering invariants (RX-DSP stop on keydown -> wire MOX -> TX-DSP start; keyup TX-DSP off -> mox_delay -> clear MOX -> ptt_out_delay -> RX-DSP restart). No direct port — C# is study-only by license. |
| Console\console.cs (TR delays)             | src/tx/TrSequencing.{h,cpp} (LIVE)                  | [DONE] | Lyra-native shipped with operator's Thetis-DB-verified defaults (mox=15/rf=50/space_mox=13/ptt_out=5/key_up=10 ms).                                                                                                                                                                    |
| Console\console.cs::m_bATTonTX             | src/tx/AttOnTxPolicy.{h,cpp} (LIVE)                 | [WIP]  | Lyra-native shipped on main. Reference posture: keydown save per-band step-att + force-31 (when PS-A off OR CW); keyup restore. Implementation verified against console.cs:30293-30327 + :30391-30410.                                                                                 |
| Console\PSForm.cs                          | src/ui/PsDialog.{h,cpp} (NEW, v0.3)                 | [PEND] | v0.3 PureSignal UI. C# study only; write Qt/QML native.                                                                                                                                                                                                                                |
| Console\HPSDR\IoBoardHL2.cs                | src/hl2_stream.cpp (LIVE)                           | [WIP]  | HL2 I/O quirks already implemented Lyra-native via the protocol-byte verification + gateway-RTL ground-truth pass (CLAUDE.md §15.26).                                                                                                                                                  |

## STAGE INDEX — ordered work plan

### P0 verbatim-rewrite arc (2026-06-09 -> current; the ACTIVE plan)

After the operator rejected the 2026-06-09 "rule-#8" retrofit deviations ("I said PORT!"), every TX-porting file is being rewritten VERBATIM (byte-faithful; only documented packaging differences — lyra::wire namespace, PORT-expands-empty, the wdspcalls seam for the statically-linked WDSP exports). Each step ships with a mechanical diff vs the reference + an operator HL2 bench gate.

| Step        | Status      | What                                                                                                          | Commit / gate      |
|-------------|-------------|---------------------------------------------------------------------------------------------------------------|--------------------|
| <b>P0.a</b> | [DONE] DONE | wire/wdspcalls.{h,cpp} — the single approved WDSP linkage seam (exact export names, PS entry family complete) | e0c0f4a 2026-06-09 |



| Step        | Status        | What                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Commit / gate                                                                                                                                                                                                                |
|-------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P0.b</b> | [DONE] DONE   | ilv.{h,c} verbatim -> wire/ILV.{h,cpp} + new wire/cmcomm.{h,cpp}                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1f49d98 2026-06-11; test_ilv ALL PASS                                                                                                                                                                                        |
| <b>P0.c</b> | [DONE] DONE   | aamix.{h,c} verbatim -> wire/AAMix.{h,cpp} + wire/resample.h RESAMPLE ABI                                                                                                                                                                                                                                                                                                                                                                                                                                                              | e0f2584; [DONE] operator HL2 RX bench PASSED 2026-06-11                                                                                                                                                                      |
| <b>P0.d</b> | [DONE] DONE   | cmbuffs/cmaster/cmsetup verbatim -> wire/CmBuffs/wire/CMaster/wire/cmsetup; full _cmaster struct (PS surface); TxChannel carve-out DELETED                                                                                                                                                                                                                                                                                                                                                                                             | afc7950 2026-06-12; [DONE] <b>operator HL2 RX-regression bench PASSED 2026-06-12</b>                                                                                                                                         |
| <b>P1</b>   | [WIP] SHIPPED | obbuffs.c verbatim -> wire/ObBuffs.{h,cpp} (the TX-out seam; separate TU from cmbuffs; sendOutbound call DEFERRED to P2)                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-06-12; dormant/wire-inert — optional operator RX smoke; clean build 0 warnings; mechanical diff IDENTICAL                                                                                                               |
| <b>P2</b>   | [WIP]         | sendOutbound / sendProtocol1Samples fidelity audit of the dormant wire layer. <b>P2.a SHIPPED 2026-06-12</b> — eer completion (verbatim struct + seam entries + the four CMaster restore sites); kills the null-peer landmine both reference functions deref. NEXT: P2.b prn outbound surface verbatim (outLRbufp/outIQbufp/OutBufp -> reference pointer fields + the HANDLE semaphore quartet; callers bend) -> P2.c wire/Network.cpp sendOutbound verbatim (P1 branch live, ETH branch DEFERRED) + restore the ObBuffs ob_main call. | Audit finding: Lyra's Step-14 OutboundRing/Ep2SendThread are functionally-parallel idiom translations (vector buffers, bool+cv semaphores) predating the verbatim mandate — retired at P4 when the verbatim chain goes live. |
| <b>P3</b>   | [PEND]        | netInterface outbound registrations (SendpOutboundTx(OutBound) etc.) — MUST register AFTER create_xmtr (verbatim setters have NO null guards)                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                              |
| <b>P4</b>   | [PEND]        | Wire-LIVE switchover                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ONE commit, full HL2 bench gate                                                                                                                                                                                              |

PureSignal is a committed feature — the P0.d \_cmaster struct carries the full PS surface so v0.3 lands on top without a retrofit.

### *Historical stage index (pre-P0 lettered stages — superseded)*

The original lettered plan below is retained for history. The P0 arc absorbed/superseded it: Stage C (ilv) = P0.b; Stage D (xcmaster) = P0.d; Stage E (TxChannel audit) = obsoleted by the P0.d TxChannel deletion; Stage B's idiom-translated AAMix was re-reported verbatim at P0.c.

| Stage    | Status                | What                                                   | Notes                                                                |
|----------|-----------------------|--------------------------------------------------------|----------------------------------------------------------------------|
| <b>A</b> | [DONE] DONE           | CMaster.cpp wire-up stubs (Send*Outbound* + SetTCIRun) | Superseded by the P0.d full verbatim CMaster port.                   |
| <b>B</b> | [DONE] DONE           | aamix.c direct port + wire into RX path                | Shipped 2026-06-08 (idiom-translated); re-reported VERBATIM at P0.c. |
| <b>C</b> | [DONE] DONE (as P0.b) | ilv.c direct port (TX I/Q interleaver)                 | Landed verbatim at wire/ILV.{h,cpp}.                                 |



| Stage    | Status                | What                                                | Notes                                                                                                         |
|----------|-----------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>D</b> | [DONE] DONE (as P0.d) | xcmaster pump body direct port                      | Landed verbatim in wire/CMaster.cpp (real stype/txid/chid + update[] CS).                                     |
| <b>E</b> | [N/A] OBSOLETE        | TxChannel reference-parity audit + reconciliation   | TxChannel DELETED at P0.d — the verbatim create_xmtr replaces it.                                             |
| <b>F</b> | [PEND] TBD            | WDSP TXA chain components per CLAUDE.md §4.1 v0.2.1 | compress / cfcomp / eqp / wcpagc — operator-driven per-feature direct port vs cffi-only call. Order TBD.      |
| <b>G</b> | [PEND] TBD            | PureSignal direct ports (v0.3)                      | iqc / calcc / xbuilder / delay. Lands when v0.3 PS work starts. Needs operator HL2+ with the PS hardware mod. |
| <b>H</b> | [PEND] TBD            | TX/PS auto-attenuator FSM port                      | C# study only (PSForm.cs); Lyra-native rewrite.                                                               |

**Ordering note:** the P0->P4 sequence above is the active linear plan. F–H land after P4 Wire-LIVE. The methodology (smallest revertible step + mechanical diff + operator bench gate) applies within each step.

## OPEN QUESTIONS FOR OPERATOR

These are decisions that shape the plan but I cannot make unilaterally — surface them when picking up tomorrow:

1. \*\*Stage C (ilv.c) vs Stage D (xcmaster) vs Stage E

(TxChannel audit) — which first?\*\* All three are defensible:

- Stage C unlocks reference-faithful TX I/Q flow.
- Stage D removes the Lyra-native pump (cleaner end-state).
- Stage E formalizes what's already shipped (lowest risk,

highest confidence-gain).

2. \*\*Do we re-port networkproto1.c + network.c + the Step 14

wire pieces direct from reference, OR leave them as the current mix of Lyra-native rewrites + original h12\_stream.cpp\*\* The Step 14 plan was midway through these when Stage B started. Direct-port would replace the rewrites with reference-verbatim ports. Operator call.

3. \*\*WDSP TXA-chain components — direct port (~150-600 LOC

each) vs cffi-only\*\* The DLL exports work; cffi is faster to ship. Direct port gives Lyra-cpp full source visibility + lets us patch behavior at the source level. Per-component operator decision; default = cffi-only unless a real source- level need surfaces.

4. \*\*PS work timing — start v0.3 (iqc/calcc/etc.) now in

parallel with TX, or sequence after TX is fully done\*\* The CLAUDE.md plan defers PS to v0.3 (after v0.2 TX release). Operator may want to start the ports early.

5. \*\*The §15.27 "opt-in toggles that become defaults" cleanup

from the Python project — does it have a lyra-cpp analog here\*\* The Python project had a long backlog of legacy toggles to retire. Lyra-cpp doesn't carry the same legacy, but new direct-port commits may obsolete some Settings surfaces operator-knows-about — track per-component.

## SHIPPED-WORK LOG

Newest at top. Cross-references SESSION\_LOG.md for full arc details.

## 2026-06-12 — P0.d cmbuffs/cmsetup VERBATIM port ([DONE] HL2 bench PASSED)

- **P0.d** afc7950 on tx-rebuild: NEW wire/cmsetup.{h,cpp}

(cmMAX\* 16/4/4/2/32 + the 8 id helpers + SetRadioStructure + set\_cmdefault\_rates); wire/CmBuffs.{h,cpp} rewritten verbatim (cmb, \*CMB, malloc0/\_aligned\_free; Phase-C retrofit deviations removed); wire/CMaster.{h,cpp} rewritten verbatim (FULL \_cmaster struct incl. the PureSignal surface, enum AudioCODEC, raw TCI fn ptrs, verbatim create/destroy\_cmater + create/destroy\_xmtr + xcmaster + all Sendp\*/Set\* setters).

- **TxChannel RAIL carve-out DELETED** (src/wdsp/TxChannel.{h,cpp}

retired) — the verbatim create\_xmtr opens the WDSP TXA channel + TX analyzer (disp 1) + out[0..2] + ILV itself through the wdspcalls seam.

- main.cpp: SetRadioStructure(2,1,1,1,0,...) config before

create\_cmater (chid(0,0)=0 RX1 / chid(1,0)=1 TX = the live layout); create\_xmtr() post-resolve\_wdsp\_calls (documented DEFERRED-CALLSITE accommodation); handler-1.5 = gated destroy\_xmtr().

- Verification: clean build ZERO warnings; test\_ilv ALL PASS;

mechanical diffs IDENTICAL (CmBuffs.{h,cpp} / cmsetup.h / CMaster.h / all 10 fully-verbatim cmaster.c functions) + live-line-subset PASS for the partially-deferred bodies (sole accommodation: (char\*) "" at XCreateAnalyzer).

- New at startup: TWO cm\_main pump threads (streams 0+1; stream 0

idles — no Inbound producer); TX stays wire-quiescent (OutboundTx null until P3).

- **[DONE] GATE PASSED 2026-06-12: operator HL2 RX-regression bench**

("RX still working") — P1 (obbuffs.c) is unblocked.

## 2026-06-09 -> 2026-06-11 — P0.a/P0.b/P0.c verbatim-rewrite arc

- Operator rejected the 2026-06-09 "rule-#8" retrofit deviations

("I said PORT!") -> every ported file rewritten VERBATIM.

- **P0.a** e0c0f4a: wire/wdspcalls.{h,cpp} — the single

approved WDSP linkage seam (fn-ptr table, exact export names, PureSignal entry family complete + dumpbin-verified). scratch/\_typedef\_fidelity\_test.cpp disproved the claimed "C++ collisions" (twin typedef / complex[2] / raw fn ptr all compile verbatim).

- **P0.b** 1f49d98: ilv.{h,c} verbatim -> wire/ILV.{h,cpp};

new wire/cmcomm.{h,cpp} (complex/PORT/malloc0/PI); pilv[] bank gone; SendpOutboundTx raw fn ptr (register AFTER create\_xmtr — no null guards, reference ordering). Mechanical diff whitespace-only; test\_ilv ALL PASS.

- **P0.c** e0f2584: aamix.{h,c} verbatim -> wire/AAMix.{h,cpp}

(supersedes the Stage-B idiom translation); wire/resample.h = public RESAMPLE ABI; wdspcalls retyped + flush\_resample; WdspEngine outbound lambda -> static fn + TU-scope context ptr. **[DONE] Operator HL2 RX bench PASSED 2026-06-11.**

## 2026-06-08 EOD — Stage B aamix.c port COMPLETE

- **Stage B (aamix.c)** [DONE] direct port shipped + bench-validated

- Commits 27ac2fa -> 7a50b07 on tx-rebuild

- Operator HL2+ bench 19:27:49 -> 19:28:47: audible RX through

both AK4951 jack and PC-Soundcard sinks; mute/unmute via dispatch; output-device swap; AGC mode changes; chain out=N tracks 1:1 with dispatch calls=N across both sinks

- B.6.b root cause: Stage-A SendpOutboundRx no-op stub at

RadioNet.cpp:272 clobbered aaMix\_ -> Outbound ~1 ms after WdspEngine::openRx1 wired the real lambda. Fixed by deleting the stub (commit 8b8e0da).

- 2-stage diagnostic instrumentation (chain @ 225518c +

dispatch @ 12369bc) localized the defect to a single line of code on the 2nd bench — pure-speculation cost avoided.

- Methodology lessons locked into this doc § "LOCKED

METHODOLOGY" above (counters-first when symptom is "nothing happens"; operator-empirical rule held; "do as Thetis does" continues to pay).

- Tasks #130 (parent) + #132 (Stage B.6.b) closed.
- Backup \_backups/lyra-cpp-2026-06-08-aamix-stage-B-COMPLETE-with-docs.bundle.

### **Earlier work**

See SESSION\_LOG.md for the full session-by-session log. The Step 14 wire-layer arc (Stages 1, 1.5, 2a, 2b1, 2b2) is covered in detail there + in EXECUTION\_PLAN.md's progress- tracking sections. TX-1 components 2-6 (main branch) are covered in project\_lyra\_cpp\_tx.md (memory file).

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## **RESUME POINTER (next session)**

### **First actions (locked discipline):**

1. `git fetch origin`
2. `git status + git log --oneline -5 tx-rebuild origin/main`
3. Read this doc + the "READ FIRST" block in `project_lyra_cpp_tx.md`
4. Read SESSION\_LOG.md newest entry for arc detail

**Then:** wait for operator decision on which Stage (C / D / E / F / open-questions item) to land next. Each Stage gets its own grounded design + reference-read + smallest-revertable- step + bench-gated implementation pattern matching the Stage B arc.

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\*Last updated: 2026-06-08 EOD — Stage B aamix.c port shipped + bench-validated. Next port target = operator's call.\*